Segment-3

Indexing & Hashing

Indexing:

→ Indexing is used to optimize the performance of a database by minimizing the number of disk accesses required when a query is processed.

→ It is a type of data structure which is used to located and access the data in a database table quickly.

Types of index:

(i) Primary Index
(ii) Secondary Index
(iii) Cluster Index

Primary Index: If the index is created on the basis of the primary key table. 3 types of Primary index.

(i) Dense index
(ii) Sparse index
(iii) Multilevel index.

■ Dense index: A dense index is an indexing where every possible search key value is included in the index, along with a pointer to the actual data record.

(এখানে আমরা 1 by 1 search করবো)

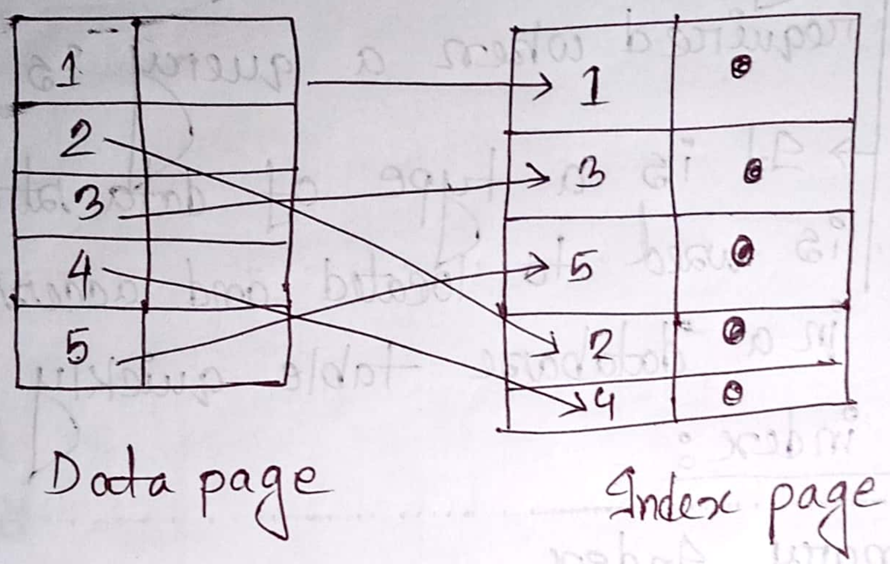
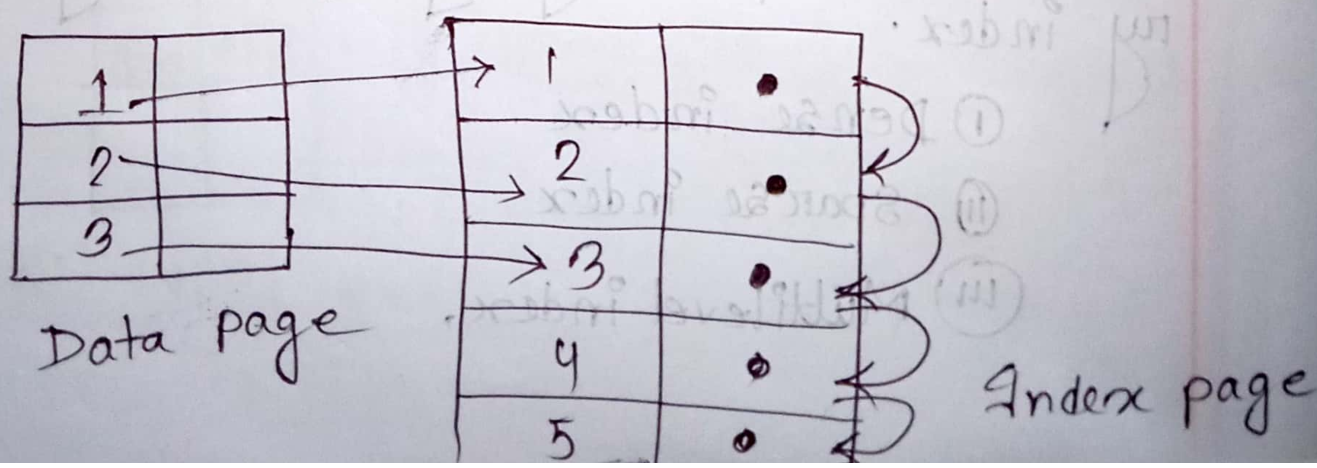


Fig : Dense index

■ Sparse index: A sparse index is an indexing where used to optimize the retrieval of records from a database table.

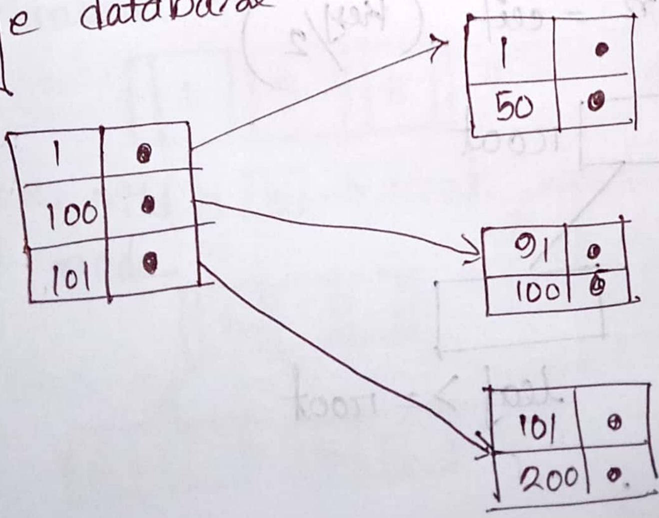
(এখানে আমরা section wise search করবে)



Difference between Dense index and sparse index :

Dense index	Sparse index
① An index record appears for every search key value in file.	① An index record are created only for some of the records.
② Indicies are faster.	② Indicies are slower.
③ Indicies requires more space	③ Indicies requries less space.
④ Indicies impose more maintainance for insertion and deletions.	④ Indicies impose less maintainance for insertion & deletions.

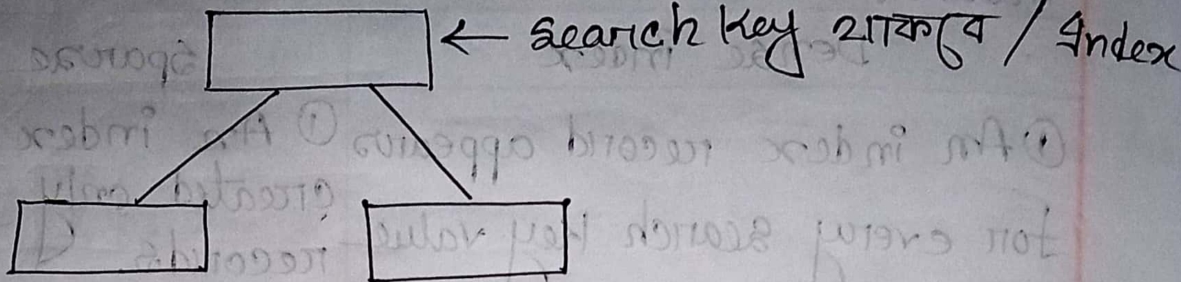
Multilevel Index: A multilevel Index also known as a ~~be~~ hierarchical index, is a data DS to improve the efficiency of data retrieval in large database.



Multilevel এর কাজ অনেকটা sparse index এর মতো।

of transaction :

B+ Tree:



Rules:

→ All data will be on leaf node or database page will be leaf node.

→ Internal node/root node stores keys

→ The leaf node are linked using linked list.

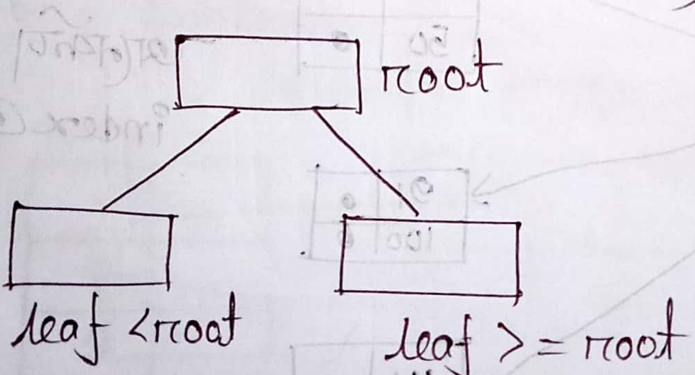
→ Data of internal node will not be repeated.

→ If order = n, then children = n and keys = n - 1

fill factor / minimum

children = ceil (key/2)

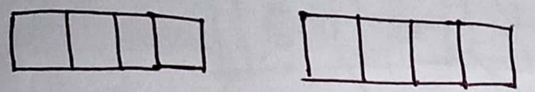
→



→ Sorting order.

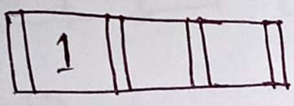
□ Insert 1, 3, 5, 7, 9, 2, 4, 6, 8, 10, 14

⇒ Order = 4, Children = 4, Keys = 4 - 1 = 3



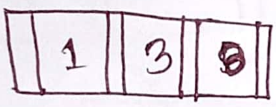
fill factor = ceil (4/2)
= 2 (50%)

step-1:

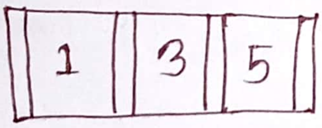


~~Inter~~ Insert-1

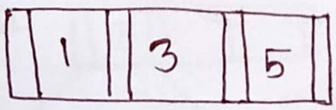
step-2:
insert - 3



step-3:
insert - 5



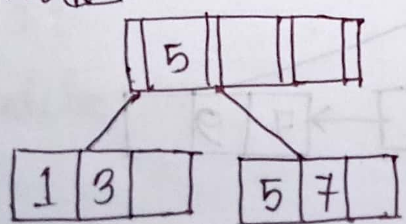
step-4:
insert - 7



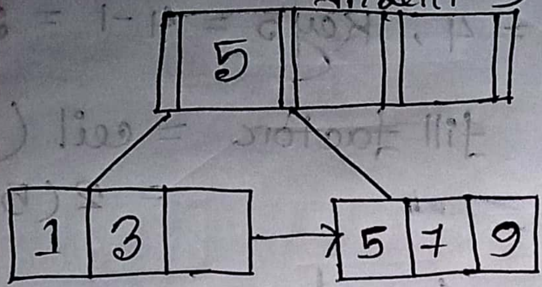
7

এখানে overflow হয়ে যাচ্ছে। তাই 7 insert হবে না।

Here, Mid = $\boxed{5}$ → root.
root node

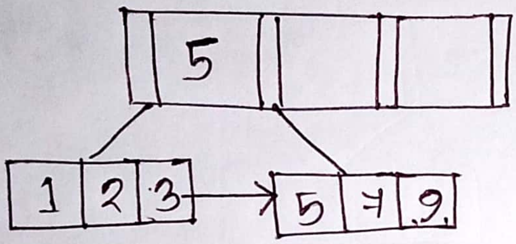


Step-5: Insert-9



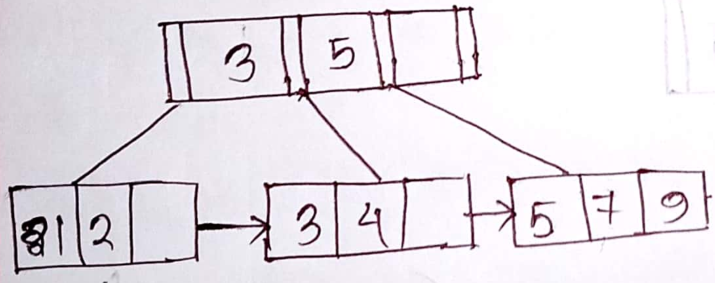
Step-6

Insert 2



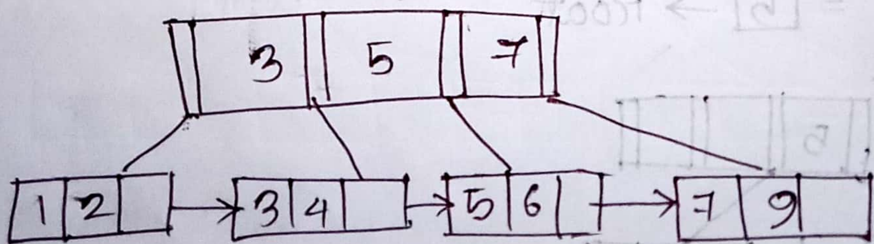
Step-7

Insert 4



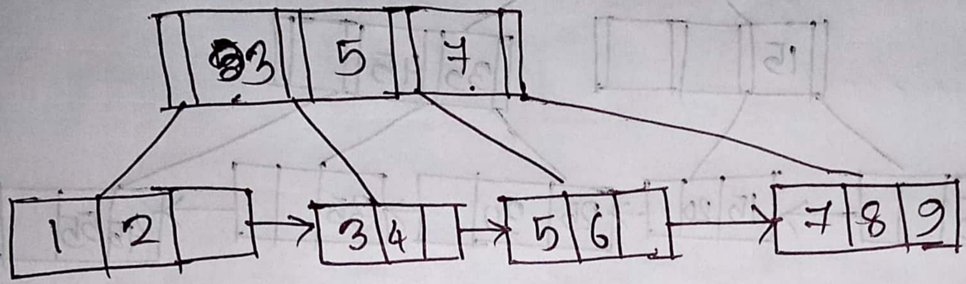
Step-8

Insert-6



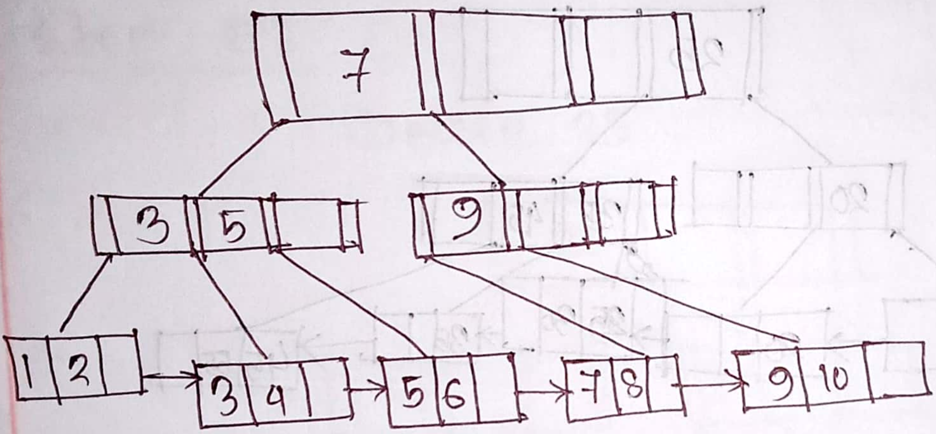
Step-9

Insert 8



Step-10

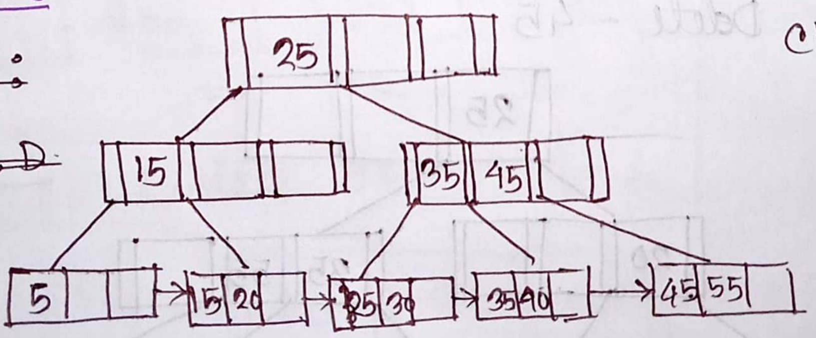
Insert 10



Deletion :

~~Step-1 :~~

~~Del D~~



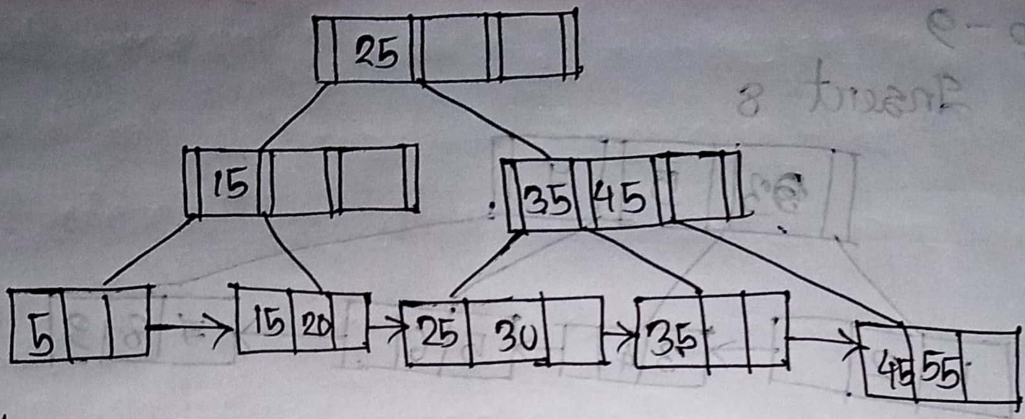
Order = 4

Children = 4

Keys = 4-1
= 3

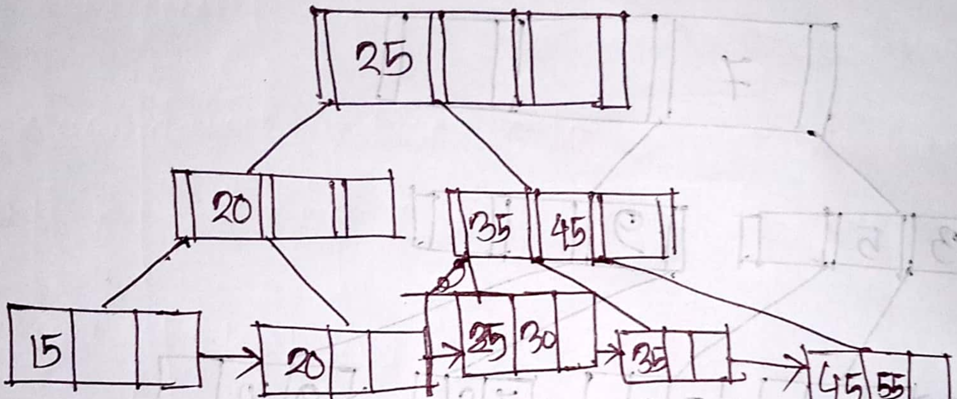
Step-1 :

Delete-40



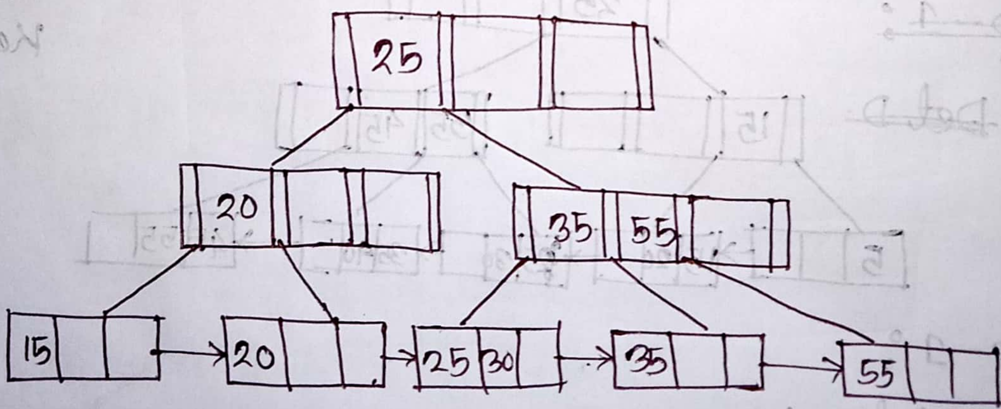
step-2 :

Delete 5



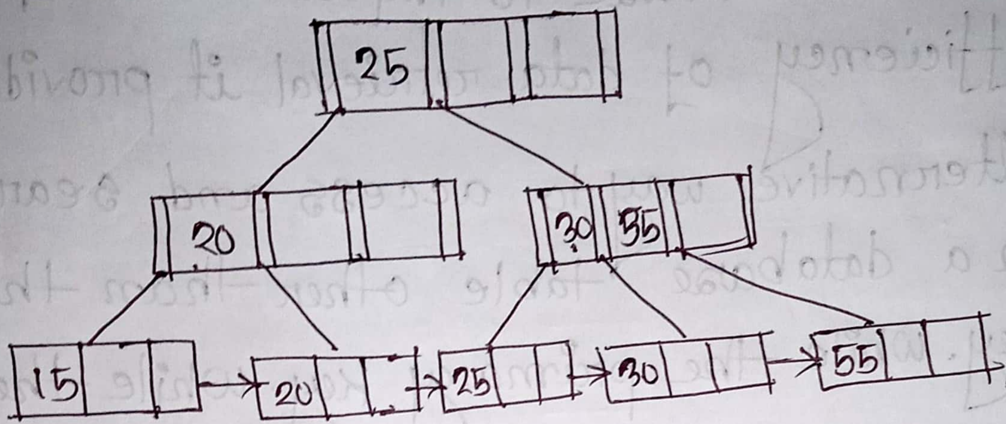
step-3 :

Delete - 45



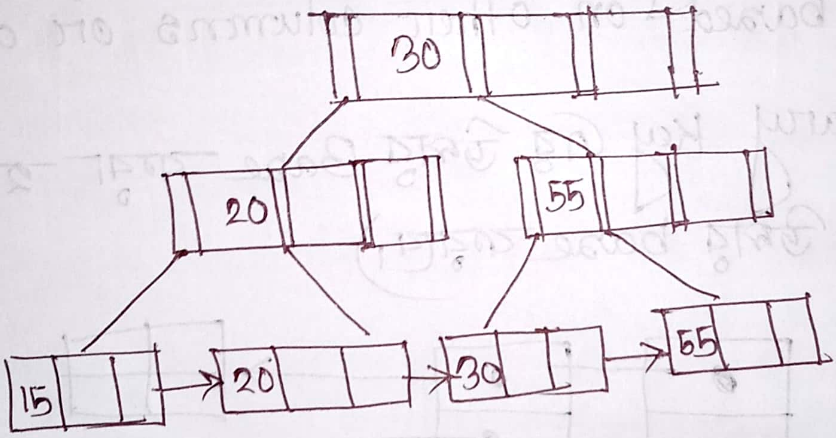
step-4:

Delete 35



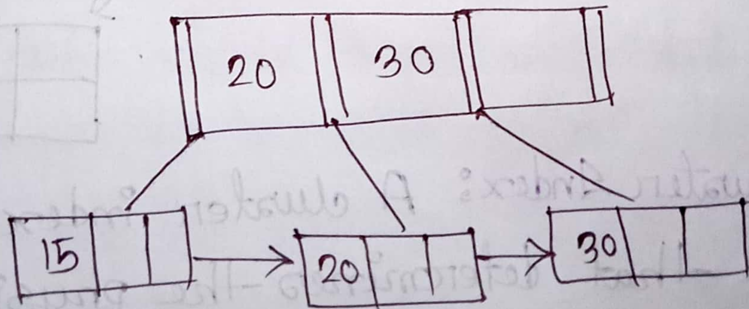
step-5:

Delete 25



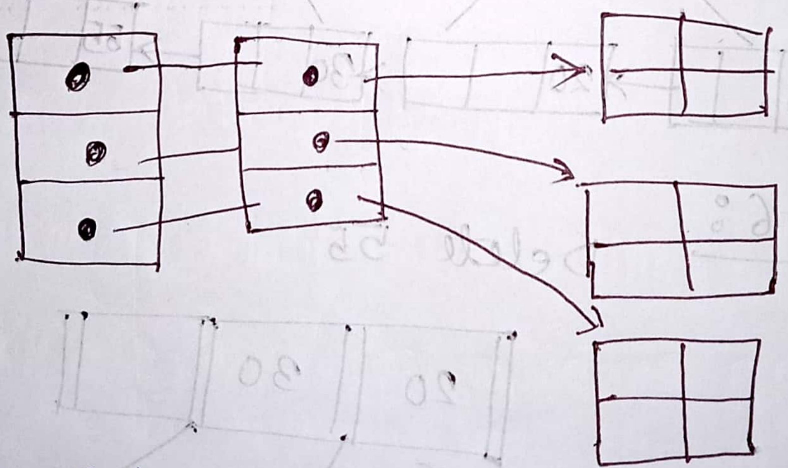
step-6:

Delete 55



Secondary Index: A secondary index used in database to improve the speed and efficiency of data retrieval it provides an alternative way to access and search for data in a database table other than the primary key. While the primary key. While the primary key uniquely identifies each row in a table, a secondary index allows for efficient searching based on other columns or attributes.

(primary key এর উপর Base করা হবেনা Candidate Key এর উপর base করবে।)



Cluster Index: A cluster index is a type of index that determines the physical order of data rows in a table. (অনেকগুলো search কে মিলায় cluster)

IV Difference between primary index & secondary index:

Primary index	Secondary index
(i) Primary index is mandatory.	(i) Secondary index is optional.
(ii) Only one for the table	(ii) Max 32 for a table
(iii) Physical mechanism effect data distribution.	(iii) Logical mechanism doesn't provide access path to the row
(iv) Can't be drop	(iv) Can be drop
(v) Provides access path to the row.	(v) Does not provide
(vi) Storage and retrieval	(vi) Only retrieval
(vii) Single Amp operation	(vii) 2 or many Amp Operation.

III Hashing:

Hashing is a technique used to quickly located a data record in a large database. It involves applying a hash function to a data item such as a key or an attribute, to generate a fixed-size value, which is typically an index or address in a data structure like a hash table.

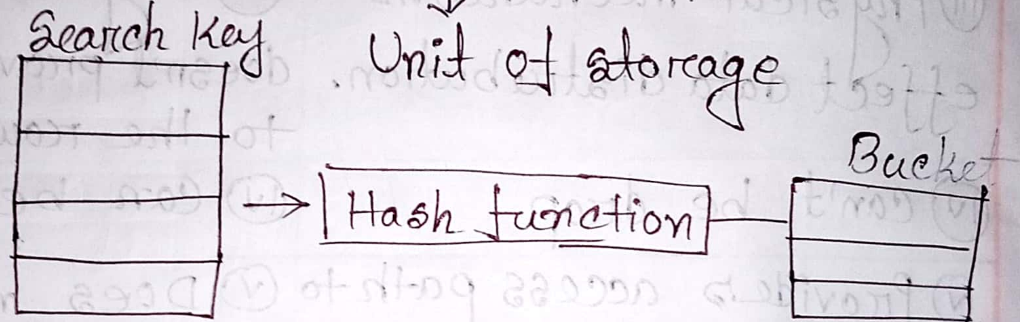
Divide function: The hashing divide function is not a standard term or concept.

$$\text{Hash Value} = \text{Key} \,\%\, \text{Table size}$$

Hashing
- Hash function
- Keys
- Bucket / data block

↓

Unit of storage



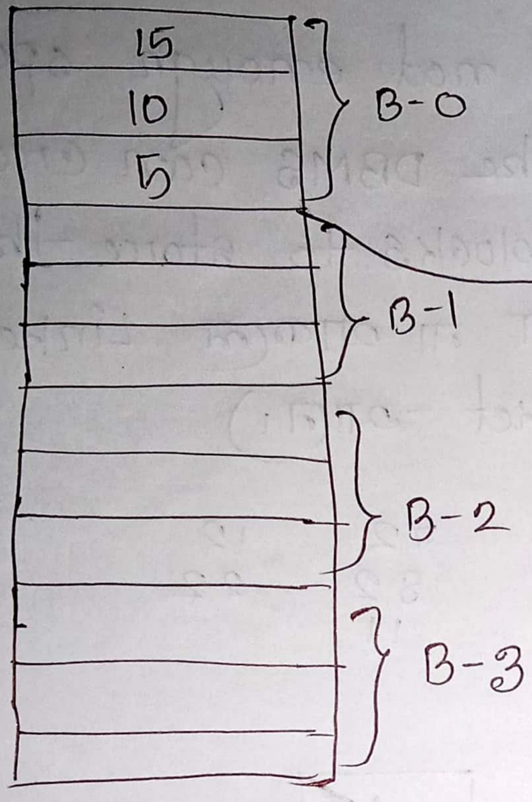
Types of Hashing:

(i) Static Hashing
(ii) Dynamic Hashing.

Static Hashing: static hashing the size of the hash table is fixed in advance and does not change as the number of elements stored in it varies

Hash table = 5   Keys = 15   address = 15 % 5
= 0

Bucket



15%5 = 0
10%5 = 0
5%5 = 0
20%5 = 0
25%5 = 0

Bucket overflow
↓ (Condition Collision)
Allocation of a few more bucket than required that time bucket overflow can occure.

Bucket overflow 2টা কারণে হয়।

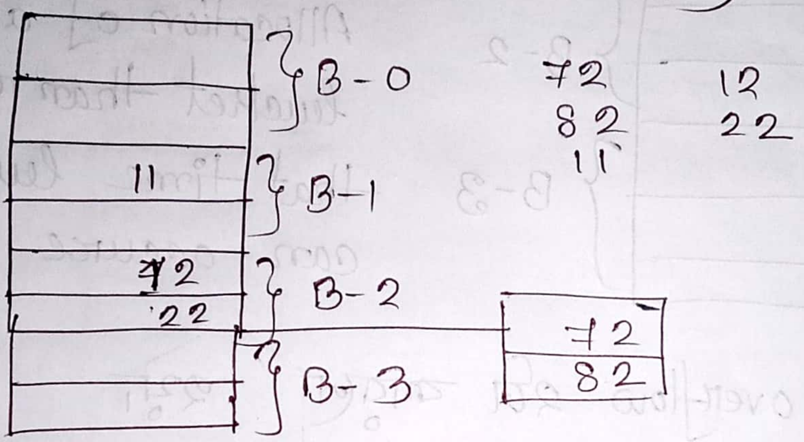
① Insufficient (Space থাকবে না কোনো ও)

ⅱ Skew (Bucket খালি নাই কিন্তু অন্য Bucket গুলো খালি থাকার পরও আমরা বলতে পারি Bucket overflow)

□ Bucket overflow solution:

- Open hashing / Linear Probing
- Close hashing / Overflow chaining

Overflow chaining: When data is inserted into a Bucket and there's not enough space in the original data block, the DBMS can creat a chain of overflow blocks to store the extra data. (Bucket এ জায়গা না থাকলে Linked List করে আরেকটা Bucket আনে।)



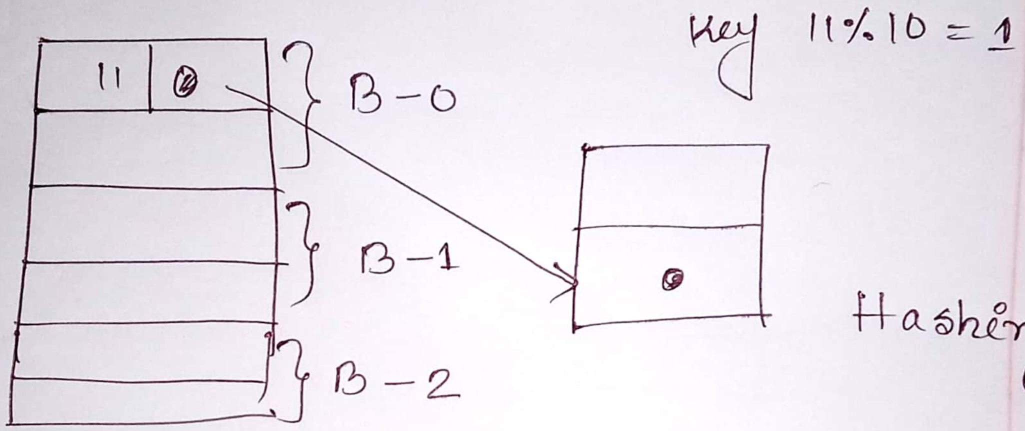
Open hashing: It is also known as closed addressing or separate chaining, is a technique used in DBMS to handle collisions in hash tables. (Bucket Limited থাকবে। আর কোনো Bucket আনতে পারবো না। এক Bucket full হয়ে গেলে অন্য Bucket এ গিয়ে search করবো।)

Dynamic: It is a technique used in Database management systems to handle the problem of managing a hash table when the number of elements (key-value pairs) in the table is not fixed and can change over time.

(Bucket বাড়তে ও পারে অথবা কমতেও পারে।)

Hashing Index:

A hash index organizes the search keys, with their association pointers into a hash file structure.



Hashing Index

Ordered Index: An ordered index also known as a sorted index is a data structure used in DBMS to improve the efficiency of searching for and retrieving records from a database.

In ordered index the data entries stored in a specific order, such as ascending or descending order, based on the values of one or more columns.

Hash Functions:

Follow are some known hashing-algorithms used in the database.

Division Method: A hash function which uses division method is represented as:

$$H(K) = K \pmod{m} \quad \text{or} \quad H(K) = K \pmod{m} + 1$$

Mid-square method: $H(K) = 1$ we square hash key (K) and eliminate digits from both ends of K^2.

Folding Method: Chop a hash key into no. of parts and compute a hash address after adding these parts and ignoring the carry.

$$H(K) = K_1 + K_2 + \cdots + K_n$$